

BCI project procedure – Sensorimotor rhythms (ERD/ERS)

EVENT-RELATED DESYNCHRONIZATION AND SYNCHRONIZATION (ERD/ERS)

Event-related desynchronization (ERD) and **event-related synchronization (ERS)** are neural oscillatory phenomena observed in EEG signals, particularly in the **mu (8-13 Hz)** and **beta (13-30 Hz)** frequency bands. ERD reflects a decrease in rhythmic brain activity linked to cortical activation, while ERS indicates a subsequent increase associated with inhibition of idling states. They both represent dynamical states of thalamocortical networks associated with cortical information-processing changes.

During **motor execution** (e.g., hand movement), **ERD arises contralaterally** (e.g., left hemisphere for right-hand movement) in sensorimotor areas before and during the movement, followed by a **bilateral beta-band ERS** (“beta rebound”) post-movement. For example, right-hand movement execution generates contralateral ERD in the left motor cortex, with ipsilateral beta ERS in the right hemisphere to suppress unrelated activity.

Motor imagery (mental simulation of movement) elicits similar but weaker ERD/ERS patterns. Kinesthetic motor imagery activates overlapping neural networks as actual movement, producing contralateral ERD in the mu/beta bands, though with delayed or reduced magnitude compared to execution. Notably, amyotrophic lateral sclerosis (ALS) patients show diminished EDF during motor imagery, highlighting shared mechanisms between imagery and execution.

Computation of the ERD

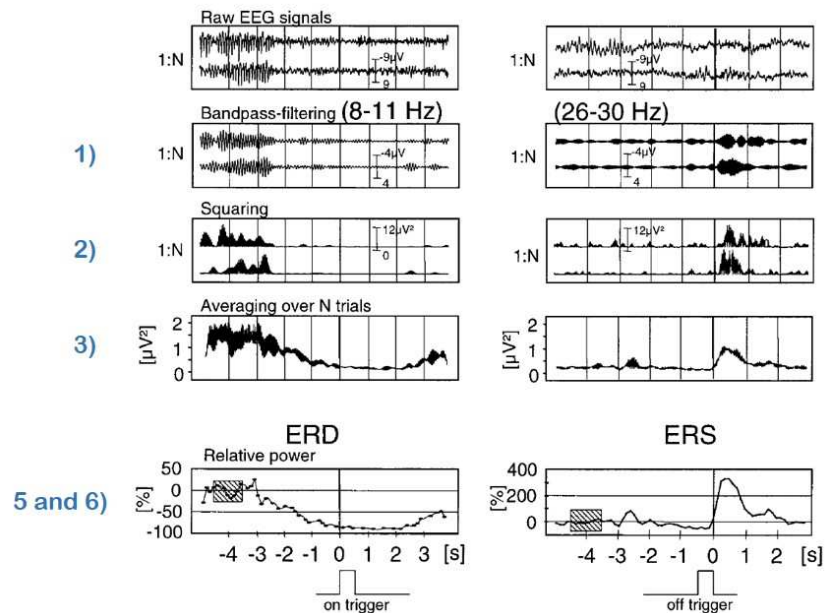
For the calculation of the ERD, all preprocessing steps prior to trial extraction are remain fully applicable: decimation, filtering, bad channel rejection, re-referencing to the common average (CAR), independent component analysis (ICA), and interpolation of bad channels.

Please follow the methodology described in *Pfurtscheller and Lopes da Silva, 1999 – Event-related EEG/MEG synchronization and desynchronization: basic principles*, available on Atenea. These steps should be applied separately for each channel, movement class, and subject.

- 1) Band-pass filter all the continuous EEG data (run).
- 2) Focus on two sensorimotor frequency bands: mu (8-13 Hz) and beta (13-30 Hz).
- 3) Construct the trials using the movement onset sample (either the one obtained in the previous (optional) step or the value from the 3rd column of the “events” field in the MAT file) along with the corresponding movement class event code. Each trial should span from -2.5 to 5.5 seconds relative to the movement onset, which is considered 0. If you’re working with a 3D matrix (recommended), you should end up with a matrix of size 60 trials x 61 channels x 4096 samples for each subject and movement class. A convenient approach is to initialize this 3D matrix with NaN values to reserve space. This also allows you to mark bad trails and bad channels using NaN, keeping everything consistent and easy to handle later.

- 4) For each channel, compute the squared amplitude of each trial to obtain the energy.
- 5) Average of power across all trials.
- 6) Smooth the power signal by averaging over time using a moving average window of 150ms. This can be implemented with the Matlab function called “movmean.m”.
The moving average is a simple low-pass FIR (Finite Impulse Response) filter commonly used to smooth sampled data. It processes M samples at a time, computing their average to produce a single output point.
- 7) Determine the reference energy for the mu and beta bands by averaging the power within the baseline interval [-1.5 s to -1 s], where 0 marks the movement onset.
- 8) Calculate the ERD as the percentage change in power relative to the reference value, indicating a power decrease (desynchronization) or increase (synchronization):

$$ERD_{channel j} = 100 \cdot \frac{Act_{channel j} - Ref_{channel j}}{Ref_{channel j}}$$



(OPTIONAL) Explore alternative methodologies for analyzing ERD/ERS activity, such as wavelet-based time-frequency analysis (<https://www.nocions.org/eeg-cwt>) or intertrial variance analysis (Shahlaei, 2018).

- Compute the grand-mean average ERD for the mu and beta bands (i.e., the mean across subjects for each EEG channel and movement class).

It may be interesting to compare the individual average ERD values for each subject with the grand-average ERD, as well as to contrast ERD responses between movement classes.

- Identify the minimum value of the grand-average ERD in both mu and beta bands.

Note: This minimum typically occurs around movement onset ($t = 0$ s). To obtain a robust estimate, calculate the minimum ERD within a time window (-100 to +100ms) relative to movement onset.

- Plot topographical maps of ERD activity based on the minimum values calculated in the previous step for each EEG electrode.
- Define a Region of Interest (ROI) for each sensorimotor rhythm by selecting the electrodes corresponding to the lowest 10% (first decile) of ERD values.
- Observe whether the ERD topographies are ipsilateral, bilateral, or contralateral – keeping in mind that participants are performing right-arm movements.

Deliverables

- D6. Plot the grand-average ERD time courses for both mu and beta bands within the defined ROIs. Focus on differences between movement classes, and provide a brief discussion of the observed patterns. Explore inter-subject variability, and include a discussion on post-movement event-related synchronization (ERS).
- D7. Display the average topographical ERD maps for each movement class. Reflect on whether topographical patterns could serve as a useful feature for distinguishing between movement classes.